

AI3604 Computer Vision

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Programming Assignment

I haven't fully finished the code template before it was published. That template assumes we use a 3D chessboard and seems to use SVD to solve. I made modifications on the original template.

About the datasets, initially I found most of the pictures failed the cv2 corner detection because of low resolution (naive interpolation works). However, I later found cv2 can successfully detect corners on 4 pictures if set adaptive threshold. So I used the original data.

In 2D cases, the transformation is:

$$Z_c \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \lambda & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & t_1 \\ r_{21} & r_{22} & t_2 \\ r_{31} & r_{32} & t_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \mathbf{K} \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{t} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \quad (1)$$

where Z_c is a scaling factor, K is the demanded intrinsic matrix. (u, v) are Image points and (x, y) are world points. Since we got dozens of sets of points on each plane, on each plane we solve

$$\mathbf{H} = \mathbf{K} \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{t} \end{bmatrix} \quad (2)$$

using least square method. Now let

$$\mathbf{B} = \mathbf{K}^{-T} \mathbf{K}^{-1} \equiv \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{12} & B_{22} & B_{23} \\ B_{13} & B_{23} & B_{33} \end{bmatrix} \quad (3)$$

because easily

$$\begin{aligned} \mathbf{r}_1 &= \mathbf{K}^{-1} \mathbf{h}_1 \\ \mathbf{r}_2 &= \mathbf{K}^{-1} \mathbf{h}_2 \end{aligned} \quad (4)$$

$$\begin{aligned} \mathbf{h}_1^T \mathbf{K}^{-T} \mathbf{K}^{-1} \mathbf{h}_2 &= 0 \\ \mathbf{h}_1^T \mathbf{K}^{-T} \mathbf{K}^{-1} \mathbf{h}_1 &= \mathbf{h}_2^T \mathbf{K}^{-T} \mathbf{K}^{-1} \mathbf{h}_2 = 1 \end{aligned} \quad (5)$$

B is symmetric. Now we have

$$\begin{aligned} \mathbf{h}_i^T \mathbf{B} \mathbf{h}_j &= \begin{bmatrix} \mathbf{h}_{1i} & \mathbf{h}_{2i} & \mathbf{h}_{3i} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{12} & B_{22} & B_{23} \\ B_{13} & B_{23} & B_{33} \end{bmatrix} \begin{bmatrix} \mathbf{h}_{1j} \\ \mathbf{h}_{2j} \\ \mathbf{h}_{3j} \end{bmatrix} \\ &\triangleq \mathbf{v}_{ij}^T = \begin{cases} 0, i=1, j=2 \\ 1, i=j=1, 2 \end{cases} \end{aligned} \quad (6)$$

$$\begin{bmatrix} \mathbf{v}_{11}^T \\ \mathbf{v}_{12}^T - \mathbf{v}_{22}^T \end{bmatrix} \begin{bmatrix} B_{11} \\ B_{12} \\ B_{22} \\ B_{13} \\ B_{23} \\ B_{33} \end{bmatrix} = 0 \quad (7)$$

Again we use least square to solve **B** and finally use cholesky to solve **K**.
The result, result from cv2, error are:

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Camera Calibration Matrix:
[[751.19839352   1.7067085   488.4782087 ]
 [   0.         700.30141416  336.70403175]
 [   0.           0.         1.         ]]
Camera Calibration Matrix by OpenCV:
Camera Matrix:
[[752.45363272   0.         451.92958604]
 [   0.         702.08358999  337.2473562 ]
 [   0.           0.         1.         ]]
Reprojection Error:
[1.044, 3.766, 4.374, 9.993]

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Written Assignment

a.

$$\begin{aligned}
 E(\mathbf{A}, \mathbf{T}) &= \sum_{i=1}^N \|\mathbf{Y}_i - \mathbf{A}\mathbf{X}_i - \mathbf{T}\|_2^2 \\
 &= \sum_{i=1}^N \|\mathbf{Y}_i\|_2^2 - 2\mathbf{Y}_i^T(\mathbf{A}\mathbf{X}_i + \mathbf{T}) + \|(\mathbf{A}\mathbf{X}_i + \mathbf{T})\|_2^2
 \end{aligned} \tag{8}$$

For each item in $\sum$:

$$\begin{cases} \frac{\partial E}{\partial \mathbf{A}} = -2\mathbf{Y}_i\mathbf{X}_i^T + 2(\mathbf{A}\mathbf{X}_i + \mathbf{T})\mathbf{X}_i^T \\ \frac{\partial E}{\partial \mathbf{T}} = -2\mathbf{Y}_i + 2(\mathbf{A}\mathbf{X}_i + \mathbf{T}) \end{cases} \tag{9}$$

To find maximum, we let $\frac{\partial E}{\partial \mathbf{A}} = 0$ and $\frac{\partial E}{\partial \mathbf{T}} = 0$.

$$N\mathbf{T} = \sum \mathbf{Y}_i - \mathbf{A} \sum \mathbf{X}_i \tag{10}$$

$$\mathbf{A} \sum \mathbf{X}_i\mathbf{X}_i^T - \sum \mathbf{Y}_i\mathbf{X}_i^T - \mathbf{A}\bar{\mathbf{X}} \sum \mathbf{X}_i^T + \bar{\mathbf{Y}} \sum \mathbf{X}_i^T = 0 \tag{11}$$

Simplify equation 11:

$$\mathbf{A} \sum (\mathbf{X}_i - \bar{\mathbf{X}})\mathbf{X}_i^T = \sum (\mathbf{Y}_i - \bar{\mathbf{Y}})\mathbf{X}_i^T \tag{12}$$

Since $(\sum (\mathbf{A} - \bar{\mathbf{A}})\mathbf{B}^T) = (\sum (\mathbf{A} - \bar{\mathbf{A}})(\mathbf{B} - \bar{\mathbf{B}})^T) = \mathbf{A}\mathbf{B}^T$:

$$\mathbf{A}\mathbf{X}\mathbf{X}^T = \mathbf{Y}\mathbf{X}^T \tag{13}$$

Finally

$$\begin{cases} \mathbf{A}^* = (\mathbf{Y}\mathbf{X}^T)(\mathbf{X}\mathbf{X}^T)^{-1} \\ \mathbf{T}^* = \hat{\mathbf{Y}} - \mathbf{A}^*\hat{\mathbf{X}} \end{cases} \tag{14}$$

(Well, I mistook $\hat{\mathbf{X}}$ for $\bar{\mathbf{X}}$ above. Anyway.)

Further existence of the results and the check of Hessian Matrix is too complex to be explained within required pages.

b. Intuitively, to solve for the 12 parameters in **A** and **T**, with each set of **X**, **Y** providing three equations, four sets of corresponding points are needed. However, the solution in a. dose not show this requirement because 3 sets of points are enough for least square method. Anyway, we only use least square method when we have more than 4 sets to ‘estimate’.